

UFEXIL200 mg / 100 mL
(Ciprofloxacin)**1. NAME OF THE MEDICINAL PRODUCT:** Ufexil 200 mg/ 100 mL**2. QUALITATIVE AND QUANTITATIVE COMPOSITION** (in active ingredient): Ciprofloxacin Lactate 2.55 mg equivalent to Ciprofloxacin 2 mg. Ciprofloxacin Lactate: $C_{17}H_{14}FN_3O_3 \cdot C_3H_5O_2$, 1-Cyclopropyl-6-fluoro-1,4-dihydro-4-oxo-7-(1-piperazinyl)-3-quinoline-carboxylic acid lactate.**3. PHARMACEUTICAL FORM:** Solution for Intravenous Infusion**4. CLINICAL PROPERTIES****4.1. Therapeutic Indications:** Ciprofloxacin is indicated for the treatment of infections caused by susceptible organism strains in the following conditions:**Lower respiratory tract infections:** Caused by *Escherichia coli*, *Klebsiella pneumoniae*, *Enterobacter cloacae*, *Proteus mirabilis*, *Pseudomonas aeruginosa*. It is not effective in pneumonia from pneumococcus. Ciprofloxacin is particularly effective against *Pseudomonas aeruginosa* in cystic fibrosis. Prevention of pulmonary anthrax development after possible or certain inhalation of anthrax bacillus seeds.**Skin and skin structure infections:** Caused by *Escherichia coli*, *Klebsiella pneumoniae*, *Enterobacter cloacae*, *Proteus mirabilis*, *Proteus vulgaris*, *Providencia stuartii*, *Morganella morganii*, *Citrobacter freundii*, *Pseudomonas aeruginosa*, *Staphylococcus aureus* and *Staphylococcus epidermidis* (methicillin resistant strains). In the last case, concomitant treatment with another antibiotic (e.g. rifampicin) is suggested, in order to prevent organism resistance to ciprofloxacin.**Bone and joint infections:** Caused by *Enterobacter cloacae*, *Serratia marcescens*, *Pseudomonas aeruginosa* and *Enterobacter spp.* in general.**Urinary tract infections:** Caused by *Escherichia coli*, *Klebsiella pneumoniae*, *Enterobacter cloacae*, *Serratia marcescens*, *Proteus mirabilis*, *Providencia rettgeri*, *Morganella morganii*, *Citrobacter diversus*, *Citrobacter freundii*, *Pseudomonas aeruginosa*.**Gastrointestinal tract infections:** Intra-abdominal infections caused by multiresistant gram-negative organisms, in combination with antimicrobials with anaerobic action.**Septicaemia, *endocarditis:** Infections by multiresistant gram-negative pathogenic hospital strains, endocarditis by Q fever (*Coxiella burnetii*)

*Ufexil should be only used:

- When *Pseudomonas* is considered AND patient is allergic to antipseudomonal penicillins/cephalosporins;
- For resistant organisms with no other alternative antibiotics available.

#Ufexil should not be used >24 hours post operation.

Consideration should be given to official guidance on the appropriate use of antibacterial agents.

4.2. Posology and method of administration**4.2.1. Dosage:** The dosage of intravenous ciprofloxacin is determined by the severity and type of infection, the sensitivity of the causative organism(s) and the age, weight and renal function of the patient.Adults: The dosage range for adults is 100-200mg twice daily. In patients with sepsis or neutropenia and in patients with *Pseudomonas aeruginosa* systematic infections, doses of 200-400mg every 8 hours are recommended. In cases of upper and lower urinary tract infections, doses of 100-200mg twice a day are recommended. In both upper and lower respiratory tract infections, 200-400mg twice a day. For prevention of anthrax 400mg are administered intravenously twice a day. The administration of the drug of this indication should be started as soon as possible after possible or certain inhalation. In most other infections, 200mg twice a day should be administered, except for *P. aeruginosa* infections where dosage should sometimes be higher.**Dosage adjustments:** Ciprofloxacin is excreted mostly by the kidneys, but it is also metabolized and excreted partly through the hepatic bile system and eliminated intestinally. These alternative excretion routes seem to balance the reduced renal function. Therefore, doses usually do not need to be adjusted, except for the following conditions:

1. **Reduced renal function:** Creatinine clearance <20ml/min or serum creatinine levels >3mg/100ml. Dose should be reduced to 50% of full 24hr dose.
2. **Reduced renal function and hemodialysis:** Dose as above. Dose should be administered after dialysis is complete.
3. **Reduced hepatic function:** No dosage adjustment is required.
4. **Reduced renal and hepatic function:** Normal 24 hr dose or reduction to 50% if necessary. Drug concentration monitoring is suggested.

Duration of treatment: The duration of therapy depends on the severity of the infection, clinical response and microbiological findings. For acute infections, treatment usually lasts 5-7 days. It is essential to continue therapy for at least 3 days after disappearance of the signs and symptoms of the infection. For the prevention of anthrax the total duration of treatment is 60 days. For cystic fibrosis the recommended duration of treatment is 10-14 days. Initial IV administration can be followed by oral administration of ciprofloxacin.**4.2.2. Administration:** Ciprofloxacin should be infused slowly, over 60 minutes. Slow infusion into a large vein reduces patient discomfort and risk of irritation of the vein. The solution for infusion can be infused either as it is, or after mixing with other compatible solutions for infusion (See "Incompatibilities"). Infusion bags can be used immediately.**Elderly:** Although higher ciprofloxacin serum levels are found in the elderly, no adjustment of dosage is necessary. Elderly patients should be treated with the minimum dosage according the severity of the infection and creatinine clearance.**Adolescents and Children:** As with other drugs in its class, ciprofloxacin has been shown to cause arthropathy in weight-bearing joints of immature animals. Although analysis of available safety data from ciprofloxacin use in human is unknown, its use in the paediatric patient population is not recommended, except for cases of cystic fibrosis paroxysms and prevention of pulmonary anthrax development.**Cystic fibrosis:** In adults with lower respiratory tract infections from pseudomonas the dosage is the same as for infections from *P. aeruginosa*. In children with cystic fibrosis (5-17 years of age) with acute respiratory inflammation, oral administration of 20mg/kg of body twice a day (1500mg maximum daily dosage) or intravenous administration of 10mg/kg of body weight every 8 hours is recommended (1200mg maximum daily dosage). The infusion should be administered in more than 60 min. Sequential therapy could also be used with the following dosage scheme: 10mg/kg of body weight intravenously every 8 hours (1200mg maximum daily dosage), followed by 20mg/kg of body weight orally twice a day (1500mg maximum daily dosage). The recommended duration of treatment is 10-14 days. Dosing in children with impaired renal and/or hepatic function has not been studied.**Prevention of pulmonary anthrax development after possible or certain inhalation of anthrax bacillus seed:** For this indication the evaluation of the benefits against the possible risks suggested that administration of ciprofloxacin in children is suitable at a dosage of 10mg/kg of body weight intravenously twice a day. The maximum of 400mg intravenously per dose should not be exceeded (800mg maximum daily dosage). The drug administration should start as soon as possible after possible or certain inhalation. The total duration of preventive therapy is 60 days.**4.3. Contraindications:** Ciprofloxacin is contraindicated in patients who have shown hypersensitivity to ciprofloxacin or other quinolone anti-infectives.

Ciprofloxacin is also contraindicated in patients with a history of tendonitis or tendon lesion or rupture. The use in children and teenagers is contraindicated, except for cases of cystic fibrosis paroxysms and prevention of pulmonary anthrax development. It should not be administered during pregnancy and lactation, since drug safety in these categories of patients has not been established.

4.4. Special warnings and special precautions for use:**Aortic aneurysm and dissection.**

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis). In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department. The use of ciprofloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Undesirable Effects). Treatment of these patients with ciprofloxacin should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment. Like other fluoroquinolones, tendonitis incidents following ciprofloxacin treatment have been reported, usually concerning the Achilles' tendon and occasionally causing tendon rupture. If tendonitis develops, treatment should be discontinued and the patient should be immobilized and ask for the physician's advice. Factors contributing to tendonitis are: age over 60 years, vigorous exercise and long-term corticosteroid treatment, as well as the initial recovery period of aged patients.

Patients should be warned of the possibility of pain of the Achilles's tendon (pain in the ankle and heel region). Ciprofloxacin should not be used by children, adolescents, pregnant or nursing mothers. Drug safety in these categories of patients has not been established. Ciprofloxacin administration caused lameness in immature dogs. Histopathological examination of weight-bearing joints in these dogs showed permanent damage to the cartilage. Related drugs, such as nalidixic acid, cinoxacin and norfloxacin, also caused damage to weight-bearing joint cartilage and other arthropathic symptoms to immature animals of various species (See "Contraindication").

Exacerbation of myasthenia gravis

Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Cystic fibrosis in children: Analysis of available drug related safety data for administration in children and adolescents below 18 years of age, that in majority suffered from cystic fibrosis, did not show drug related damages of cartilage and joint. Clinical and pharmacokinetic data are available to support the administration of ciprofloxacin in children with cystic fibrosis for treatment of acute respiratory inflammation due to *P. aeruginosa*. Nevertheless its use in other diseases except for cystic fibrosis is contraindicated. In patients with cystic fibrosis ciprofloxacin has caused, in rare cases, photosensitivity reactions. Therefore it is suggested that these patients should avoid prolonged exposure to sun during therapy with ciprofloxacin. Nevertheless if this is not possible the patient should use sunblock.**Musculo-skeletal system:** The risk of developing fluoroquinolone-associated tendonitis and tendon rupture is further increased in people older than 60, in those taking corticosteroid drugs, and in kidney, heart and lung transplant recipients. Patients experiencing pain, swelling, inflammation of a tendon or tendon rupture should be advised to stop taking their fluoroquinolone medication (Ciprofloxacin) and to contact their health care professional promptly about changing their antimicrobial therapy. Patients should also avoid exercise and using the affected area at the first sign of tendon pain, swelling or inflammation. **Precautions:** Like all other quinolones, ciprofloxacin can cause excitation of the Central Nervous System leading to tremor, restlessness, light-headedness, confusion and rarely illusion or convulsions. Ciprofloxacin should therefore be used with caution in patients with known or suspected CNS disorders such as severe brain arteriosclerosis or epilepsy, or in patients with other factors that predispose to convulsions (See "Adverse effects"). Caution is necessary in patients with affected hepatic or renal function, G6PD insufficiency or intense myosalgia. Exposure to light and radiation should be avoided. Ciprofloxacin crystals are rarely observed in human urine, but more often in lab animal urine. Crystaluria, related to ciprofloxacin, has been rarely reported in human since human urine is alkaline. Patients receiving ciprofloxacin should be well hydrated and urine alkalinity should be avoided. The recommended daily dose should not be exceeded. Dosage adjustment is necessary in patients with effected renal function (see "Dosage & Administration"). Pseudomembranous colitis has been observed with almost all antibacterial drugs. For this reason in patients that experience diarrhoea related to the drug use, the possibility of pseudomembranous colitis development should be examined. Colitis can be mild, severe and even life threatening. Mild conditions usually respond to discontinuation of the drug. Moderate or severe conditions need different measures.**Prolonged, disabling and potentially irreversible serious adverse drug reactions.**

Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors.

Ciprofloxacin should be discontinued immediately at the first sign or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Tendinitis and tendon rupture

Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometime bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age) with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids and in patients receiving daily doses of 1000 mg levofloxacin. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with ciprofloxacin should be discontinued and alternative treatment should be considered.

The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.

Peripheral neuropathy

Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones.

Patients under treatment with ciprofloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see section Undesirable Effects).

Psychiatric reactions

Psychiatric reactions may occur even after the first administration of fluoroquinolones, including ciprofloxacin. In rare cases, depression or psychotic reactions can progress to suicidal ideations/thoughts and self-injurious behaviour, such as attempted or completed suicide (see section 'Undesirable effects'). In the event that the patient develops these reactions, ciprofloxacin should be discontinued and appropriate measures instituted. Caution is recommended if ciprofloxacin is to be used in psychotic patients or in patients with a history of psychiatric disease.

4.5. Interaction with other medicinal products and other forms of interaction:Concurrent administration of ciprofloxacin with theophylline may lead to increased serum concentrations of theophylline and prolongation of its elimination half-life. This may lead to increased risk of theophylline-related adverse reactions. If concurrent administration cannot be avoided, serum levels of theophylline should be monitored and dosage adjustment should be made. Concurrent oral administration of ciprofloxacin with iron, sucralfate, didanosine and antiretroviral drugs containing magnesium, aluminium or calcium reduces ciprofloxacin absorption, resulting in lower serum and urine levels. Ciprofloxacin should be administered 1-2 hours prior to or at least 4 hours after these compounds. This is not necessary in the case of H₂ receptors inhibitors.

Probenecid interferes with renal tubular secretion of ciprofloxacin and produces increased serum ciprofloxacin levels. This should be considered if patients are receiving both drugs. Intravenous administration of azlocillin can cause an increase of ciprofloxacin concentration. As with other wide spectrum antibiotics, prolonged use of ciprofloxacin can result in overgrowth of non-susceptible organisms. Repeated evaluation of patient's condition and microbial susceptibility testing are essential. If superinfection occurs during therapy, appropriate measures should be taken. Animal studies have shown that combining very high doses of quinolones and certain nonsteroidal anti-inflammatory drugs (not acetylsalicylic acid) can lead to convulsions. The same effect has been observed when the drug is combined with foscarnet. Concurrent administration of ciprofloxacin with warfarin can enhance the latter's anticoagulant activity. Transiently increased serum creatinine levels have been observed on concomitant administration of ciprofloxacin and cyclosporine. Therefore, serum creatinine levels must be tested regularly (twice a week). There are signs of synergic nephrotoxicity in concurrent use with cyclosporin. In some case, concurrent administration of ciprofloxacin and glibenclamide can enhance glibenclamide activity (hypoglycemia). Opioids should not be administered during operational period in patients taking ciprofloxacin. Metoclopramide accelerates ciprofloxacin absorption, and peak plasma concentrations are achieved in less time. No effects in its bioavailability were observed. Concurrent administration with methotrexate can lead to increased methotrexate levels in plasma and increase its toxicity. Therefore, patients taking methotrexate should be monitored carefully in case of concurrent administration of ciprofloxacin.

4.6. Pregnancy and lactation:**Pregnancy:** Reproduction studies in mice and rats using doses up to 6 times higher than normal daily human dosage have shown no reduction in fertility or damage to the fetus due to ciprofloxacin. In rabbits, like most antimicrobial drugs, ciprofloxacin (30 and 100mg/kg orally) caused gastrointestinal disorder resulting in loss of weight to the mother and increase in miscarriage rate. None of the two doses showed any teratogenic effects. IV administration in doses up to 20mg/kg showed no toxicity to the mother or the fetus nor any teratogenic effects. There are, however, no well-documented studies in pregnant women. Since ciprofloxacin, like other drugs of its category, can cause arthropathy in animals whose development is not complete, its use during pregnancy is contraindicated.**Lactation:** Ciprofloxacin is excreted in human and rat milk, like other drugs of this class. For this reason and due to the possibility of serious adverse effects from ciprofloxacin to nursing infants, the discontinuation of either nursing or ciprofloxacin administration should be decided, taking into consideration the importance of the drug to the mother.**4.7. Effect on ability to drive and use machines:** Ciprofloxacin can affect the speed of reaction to such an extent that the ability to drive or to operate machinery is impaired. This applies particularly in combination with alcohol.**4.8. Undesirable effects:****Psychiatric disorders**

Rare: Depression (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide)

Very Rare: Psychotic reactions (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide)

Exacerbation of myasthenia gravis:

Post Marketing Experience.

Musculoskeletal and connective tissue disorder*

Nervous system disorders*

General disorders and administrative site conditions*

Psychiatric disorders*

Eye disorders*

Ear and labyrinth disorders*

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious adverse drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see section Warnings and Precautions).

Hepatic impairment, renal impairment, anaphylactic reactions, haemolysis and hypoglycemia (especially when concomitantly administered with glibenclamide) can occur upon quinolone administration and could be life-threatening, if not detected by the physician. Moreover, tendon inflammation (predominantly of the Achilles tendon) has been reported which may lead to tendon rupture. Patients concurrently receiving coumarin anticoagulants must be carefully monitored, because of the risk of increased anticoagulant activity. Injection site reaction associated with irritation and pain may sometimes occur. In a small number of patients this may be accompanied by phlebitis or thrombophlebitis. Local reactions at the injection site have been reported after intravenous administration of ciprofloxacin. These reactions are more frequent if the infusion time is 30 min or less. They may appear as local skin reactions, which disappear immediately after the injection. Intravenous administration is therefore not contraindicated unless reaction recur or worsen. Ciprofloxacin is generally well tolerated. In the course of clinical studies, 2799 patients received 2868 series of the drug. Adverse effects probably connected with the drug were presented in 7.3% of the series, 9.2% of the effects eventually connected with the drug and 3% of the effects were not likely related. Ciprofloxacin administration was discontinued because of adverse effects in 3.5% of series. Those concerned mostly the gastrointestinal system (1.5%), the skin (0.6%) and the Central Nervous System (0.4%). The most common effects, either related to the drug or not, were nausea (5.2%), diarrhea (2.3%), vomiting (2.2%), abdominal pain or discomfort (1.7%), headache (1.2%), restlessness (1.1%) and rash (1.1%). Adverse effects observed in every organic system are reported below. Those typical for quinolones are in italics.

Gastrointestinal: Nausea, diarrhea, vomiting, dyspepsia, abdominal pain, flatulence, anorexia. In case of severe diarrhea during or after treatment, the patient must be examined for pseudomembranous colitis. In these case ciprofloxacin must be discontinued and suitable treatment should be administered (e.g. vancomycin, 4 x 250mg a day, orally). Drugs that interfere with peristalsis should be avoided. Rarely were observed: oral ulceration, oral candidiasis, dysphagia, intestinal perforation and gastrointestinal haemorrhage.

Central nervous system: Vertigo, headaches, weakness, restlessness tremor. Very rarely: lightheadedness, perspiration, balance disorders, insomnia, nightmares, illusions, manic reactions, irritability, ataxia, convulsions, lethargy, somnolence, weakness, malaise, anorexia, phobia, depersonalization, depression, hallucinations, increased intracranial pressure, stress, confusion, painful peripheral paresthesia. In certain cases, these reactions are met after administration of the first dose. In these cases, ciprofloxacin should be discontinued and the patient should remain under observation.

Skin/Hypersensitivity: In some cases the following reactions are observed after ciprofloxacin administration. In these instances the administration of the drug should be discontinued and the physician should be informed. Skin reactions such as rash, urticaria, pharmaceutical fever.

Rarely: Petechiae, haemorrhagic blisters and vesicles, skin coloring, erythema nodosum, erythema multiforme, skin candidiasis, flushing, photosensitivity. Stevens-Johnson syndrome, Lyell syndrome. Interstitial nephritis, hepatitis, hepatic necrosis that very rarely evolves into life-threatening hepatic impairment. Anaphylactic reactions (eg. face, neck and lips edema, edema of the conjunctivae or the hands, edema of the throat, vasoneural edema, anhelation progressing to life-threatening shock, urticaria), are occasionally observed after the first administration of ciprofloxacin. In these cases, ciprofloxacin should be discontinued and suitable treatment should be administered.

Special senses: Rarely, blurred vision, visual disorders (color perception anomalies, overbrightness of lights), reduced visual acuity, diplopia, eye pain, tinnitus, transient hearing loss (especially in high frequencies), bad taste, olfactory disorders.

Musculoskeletal: Rare arthralgias or back pain, joint stiffness, joint edema, neck or chest pains, flare up of gout, muscle pains, tenosynovitis. In isolated cases, tendonitis of the Achilles' tendon was observed during ciprofloxacin administration. Also, partial or complete rupture of the tendon occurred, particularly in elderly patients who earlier received glucocorticoids regularly. For this reason, if tendonitis is suspected (painful edema), ciprofloxacin should be discontinued and the physician should be informed.

Renal / Urogenital: Rarely: interstitial nephritis, nephritis, renal insufficiency, polyuria, urine retention, vaginitis, urethral haemorrhage, acidosis.

Cardiovascular: Palpitation and rare instances of auricular flutter, ventricular ectopia, syncope, hypertension, angina pectoris, myocardial infarction, cardiopulmonary arrest, cerebral thrombosis, migraine, collapse.

Respiratory: Rarely, Epistaxis, laryngeal or pulmonary edema, hiccup, haemoptysis, dyspnoea, bronchospasm, pulmonary embolism. Most of these incidents were mild or moderate, regressed soon after the drug was discontinued and required no therapy. In some cases nausea, vomiting, tremor, restlessness and palpitation were considered by the researchers to be due to high plasma levels of theophylline, possibly due to interactions between the drug and ciprofloxacin. Prolonged or repeated administration of ciprofloxacin can lead to superinfections by resistant bacteria or fungi.

Adverse effects in laboratory tests: Variations in laboratory parameter, recorded as adverse effects, not specifically related to the drug:

Hepatic: Increase in SGPT (ALT) (1.9%), SGOT (AST) (1.7%), alkaline phosphatase (0.8%), LDH (0.4%), serum bilirubin (0.3%).

Haematologic: Eosinophilia (0.6%), leucopenia (0.4%), agranulocytemia, lower blood platelet count (0.1%), higher blood platelet count (0.1%), pancytopenia (0.1%).

Renal: Serum creatinine increase (1.1%), blood urine nitrogen increase (0.9%). Crystalluria, cylindruria and haematuria have been reported.

Other changes, reported in rare instances, were: increase in γ -GT, increased serum amylase, decrease in blood sugar, increase in uric acid, lower hemoglobin levels, anemia, increased number of large mononucleates in blood, leucocytosis, haemolytic anaemia, changes in prothrombin time.

4.9. Overdose: In the event of acute, excessive oral overdosage, reversible renal toxicity has been reported in some cases. Therefore, apart from routine emergency measures, it is recommended to monitor renal function and to administer Mg- or Ca- containing antacids which reduce the absorption of ciprofloxacin. Only a small amount of ciprofloxacin (<10%) is removed from the body after haemodialysis or peritoneal dialysis.

5. PHARMACOLOGICAL PROPERTIES: ATC code: J01MA02

5.1. Pharmacodynamic Properties: Ciprofloxacin hydrochloride is a synthetic wide spectrum antibiotic. Ciprofloxacin is a fluoroquinolone, available in the form of monohydrochloric monohydrate salt of 1-cyclopropyl-6-fluoro-1,4-dihydro-4-oxo-7-(1-piperazinyl)-3-quinoline carboxylic acid. It is a yellowish to yellow crystal powder and has a molecular weight of 385.8. Its empirical formula is $C_{18}H_{18}FN_4O_3 \cdot HCl \cdot H_2O$. *In vitro*, ciprofloxacin is effective against a wide range of gram-negative and gram positive microbes, including *P. aeruginosa*. It is also effective against gram-positive microbes such as *Staphylococcus* and *Streptococcus*. Anaerobes are generally less susceptible. Ciprofloxacin has a rapid bactericidal action, not only in the proliferation phase but also in the testing phase. During the proliferation phase of bacterium a segmental twisting and untwisting of the chromosomes take place. An enzyme called DNA gyrase plays a decisive part in this process. Ciprofloxacin inhibits this DNA gyrase in a way that arrests the bacterial metabolism, since vital information can no longer be read from the bacterial chromosome. Resistance to ciprofloxacin develops slowly and in stages (multiple-step type). Plasmid-mediated resistance development of the kind that occurs with β -lactam antibiotics, tetracyclines or aminoglycosides has not been observed with ciprofloxacin. It is of clinical interest that plasmid-carrying bacteria are also completely sensitive to ciprofloxacin. On account of its different mode of action, parallel resistance to other important, chemically different, active substance groups, such as β -lactam antibiotics, aminoglycosides, tetracyclines, macrolide or peptide antibiotics, sulphonamides, trimethoprim or nitrofurantoin derivatives generally is not seen with Ciprofloxacin. In its indication area ciprofloxacin remains completely effective on pathogens resistant to the above mentioned groups of antibiotics. Parallel resistance is observed within the group of gyrase inhibitors. However, because of the high primary sensitivity to ciprofloxacin shown by most organisms parallel resistance is less pronounced with this drug. Ciprofloxacin is thus often still effective on pathogens that are already resistant to the less effective gyrase inhibitors. Because of its chemical structure ciprofloxacin is not affected from β -lactamases. Ciprofloxacin can be used in combination with another antibiotic. In-vitro studies with usually sensitive pathogens, have shown additive action of ciprofloxacin in combination with β -lactam antibiotics and aminoglycosides. Synergy or antagonism in efficacy were rarely observed. Possible combination drugs include: against *Pseudomonas spp*: azlocillin, ceftazidime - against *Streptococci*: mezlocillin, azlocillin, other effective β -lactam antibiotics, against *Streptococci*: β -lactam antibiotics, particularly isoxazolympenicillins, vancomycin - against anaerobes: metronidazole, clindamycin. While *in vitro* studies have shown the susceptibility of most strains of the following organisms, clinical effectiveness against infections other than those in the "Indications" paragraph, has not been established.

Gram negative: *Escherichia coli*, *Klebsiella species* (including *Klebsiella pneumoniae* and *Klebsiella oxytoca*), *Enterobacter species*, *Citrobacter species*, *Edwardsiella tarda*, *Salmonella species*, *Shigella species*, *Proteus mirabilis*, *Proteus vulgaris*, *Providencia stuartii*, *Providencia rettgeri*, *Morganella morganii*, *Serratia species* (including *Serratia marcescens*), *Yersinia enterocolitica*, *Pseudomonas aeruginosa*, *Acinetobacter species*, *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Haemophilus ducreyi*, *Neisseria gonorrhoeae*, *Neisseria meningitidis*, *Branhamella catarrhalis*, *Campylobacter species*, *Aeromonas species*, *Vibrio species* (including *Vibrio cholerae*) *Brucella melitensis*, *Pasteurella multocida* and *Legionella species*. *Brucella species* susceptibility is marginal.

Gram positive: *Staphylococcus aureus* (including resistant and nonresistant to methicillin strains), negative for coagulase, *Staphylococcus species* (including *Staphylococcus epidermidis*), *Streptococcus pyogenes* and *Streptococcus pneumoniae*.

Most *Streptococcus* strains, including *Streptococcus faecalis*, are only moderately susceptible to ciprofloxacin, like *Mycobacterium tuberculosis* and *Chlamydia trachomatis*. *Staphylococci* are generally moderately or marginally susceptible. Most *Pseudomonas cepacia* strains and some *Pseudomonas maltophilia* strains are resistant to ciprofloxacin, like the anaerobe bacteria, including *Bacteroides fragilis* and *Clostridium difficile*. Ciprofloxacin is less active when tested in acidic pH. The inoculum size has little effect when tested *in vitro*. Minimal bactericidal concentration (MBC) is generally not more than double the minimal inhibitory concentration (MIC). Resistance to ciprofloxacin *in vitro* usually develops slowly (multiple-step type). Administration of ciprofloxacin after exposure to anthrax bacillus aims at decreasing the possibility of development of disease. The levels achieved after administration of ciprofloxacin in humans are thought to be sufficient for anticipated clinical success of the drug for the specific indication.

5.2 Pharmacokinetic Properties:

Absorption: Following an intravenous infusion of ciprofloxacin the mean maximum serum concentrations were achieved at the end of infusion.

Pharmacokinetics of ciprofloxacin were linear over the dose range up to 400mg administered intravenously.

Time (h)	Mean Ciprofloxacin Serum Concentrations (mg/l) after Intravenous Administration [Time from start of infusion (in hours)]		
	100mg IV (30min infusion)	200mg IV (30 min infusion)	400mg IV (60min infusion)
0.50	1.80	3.40	3.20
0.75	0.80	1.40	3.50
1.00	0.50	1.00	3.90
1.50	0.40	0.70	1.80
2.50	0.30	0.50	1.20
4.50	0.20	0.30	0.70
8.50	0.10	0.10	0.40
12.50	0.04	0.10	0.20

Distribution: The substance is present in plasma largely in a non-ionised form. Ciprofloxacin can diffuse freely into the extravascular space. The large steady-state volume of distribution of 2-3 l/kg body weight (a little lower in elderly people) shows that ciprofloxacin penetrates into tissues resulting in concentration which clearly exceed the corresponding serum levels, in men and women, especially in genital tissue. Ciprofloxacin is present in active form in saliva, nasal secretions, sputum, skin blister fluid, lymph, peritoneal fluid, bile and secretions. Although the drug diffuses inadequately into cerebrospinal fluid (CSF), CSF concentrations are generally the 10% of peak serum concentrations. Ciprofloxacin has been detected in the skin, fat, muscles, cartilages and bones. The protein binding of ciprofloxacin is 20 to 40%, which is not high enough to cause important interactions from protein-binding with other drugs.

Metabolism: Ciprofloxacin is also metabolized. Small concentration of 4 metabolites have been reported in the human urine, which all together, are equivalent to almost 35% of an oral dose. They were identified as desethyleneciprofloxacin, sulphociprofloxacin, oxociprofloxacin and formylciprofloxacin. The metabolites have antimicrobial activity but they are less active than the unchanged ciprofloxacin.

Excretion: Ciprofloxacin is largely excreted unchanged both renally and, to a smaller extent, non-renally. The elimination half-life of unchanged ciprofloxacin over a period of 24-48 hours post-dose is 3.1-5.1 hours. The elimination kinetics are linear and after repeated dosing at 12 hourly intervals, no further accumulation is detected after the distribution equilibrium is attained (at 4-5 half-lives). Renal elimination takes place mainly during the first 12 hours after dosing. In patients with renal function the half life of ciprofloxacin is slightly prolonged and dosage adjustments may be necessary (See "Dosage").

Excretion of Ciprofloxacin (% of dose)		
Intravenous administration		
	Urine	Faeces
Ciprofloxacin	61.50	15.20
Metabolites (M ₁ -M ₄)	9.50	2.60

After single dose intravenous administration, 75% is eliminated by the urine and 34% in the faeces. Only 10-20% of single dose, oral or intravenous, is eliminated in the form of metabolites. Ciprofloxacin is almost fully excreted in the urine 24 h after administration. Renal clearance of ciprofloxacin is between 0.18-0.3 l/h. kg and total body clearance between 0.48-0.60 l/h. kg. Renal clearance of ciprofloxacin exceeds the normal speed of 120ml/min of glomerular filtration. This fact suggests that active secretion by the renal tubules occurs in addition to normal glomerular filtration. Non-renal clearance of ciprofloxacin is mainly due to active transintestinal secretion as well as metabolism. 1% of the dose is via the biliary excreted route.

Cystic fibrosis in children: In paediatric cystic fibrosis patients dosages of 20mg/kg orally twice daily or 10mg/kg iv three times daily are recommended to achieve plasma concentration/time profiles comparable to those achieved in the adult population at the currently recommended dosage regimen.

6. Pharmaceutical properties

6.1. List of excipients: Lactic acid, Sodium chloride, Hydrochloric acid, Water for injections.

6.2. Incompatibilities:

Significant incompatibilities: Ciprofloxacin solution: Unless compatibility with other infusion solutions/ drugs has been confirmed, the infusion solution must always be administered separately. The visual signs of incompatibility are e.g. precipitation, clouding, and discolouration.

Incompatibility appears with all infusion solutions/drugs that are physically or chemically unstable at the pH of the solution (e.g. penicillin's, heparin solutions), especially in combination with solution adjusted to an alkaline pH (pH of the ciprofloxacin infusion solution: 3.9-4.5). No additions should be made to the mixed final ciprofloxacin suspension.

Compatible solutions for IV infusion: Ciprofloxacin is compatible with physiological saline, Ringer solution, 5% and 10% dextrose, 5% dextrose and 0.225% or 0.45% NaCl and 10% fructose. When ciprofloxacin infusion solutions are mixed with compatible infusion solutions, they should be administered shortly after admixture for microbiological and light sensitivity reasons.

6.3. Shelf Life: 36 months

6.4. Special Precautions for storage: Store at a temperature below 30°C, protected from light.

Since ciprofloxacin is photosensitive, the infusion bottles should be removed from the box only immediately before use. The solution for infusion should be protected from excessive heat and should not be refrigerated. According to good pharmaceutical practice, every aqueous solution should be used within 24h from its preparation.

6.5. Nature and contents of container: The product is packaged in polypropylene bags of 100ml placed in carton boxes of 10 bags and in polypropylene bottles of 100ml placed in carton boxes of 10 or 20 bottles.

6.6. Instruction for use and handling: See "Dosage". Ufexil should not be used after the expiry date. Keep out of reach and sight of children.

7. Manufacturer: DEMO S.A. Pharmaceutical Industry, 21st km National Road Athens-Lamia, 145 68 Krioneri Attiki, Greece.

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Date of revision of text: July 2024



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